P-Block Element

1. The maximum number of electrons that can be accommodated in a set of porbital is:

1) $3e^-$
2) $6e^-$
3) $8e^-$
4) $4e^-$

2. Boron is mainly found in nature in which form:

1) Orthoboric acid
2) Borax
3) Kernite
4) All of these

3. The correct order of atomic radii of boron family is

1) $B < Al < Ga < In < Tl$
2) $B < Ga < Al < In < Tl$
3) $B < Al < Ga < In < Tl$
4) $B > Ga > In > Al > Tl$

4. $Al_{(s)} + \text{Conc } HNO_3 \rightarrow$ ___ X _____. (X becomes passive)

1) Al_2O_3
2) $Al(NO_3)_3$
3) $Al(NO_2)_3$
4) $Al(OH)_3$

5. Given reaction is

$Na_2B_4O_7.10H_2O \xrightarrow{\Delta}$ __ X ___ $\rightarrow Y + Z$ then, X, Y and Z are:

1) $Na_2B_4O_7, NaBO_3 \& B_2O_6$
2) $Na_2B_4O_7, NaBO_2 \& B(OH)_3$
3) $Na_2B_4O_7, NaBO_2 \& B_2O_3$
4) $Na_2B_4O_7, NaBO_3 \& B_2O_3$

6. Which of the following statement is correct regarding boric acid?

1) It has a layer structure
2) It is a weak monobasic acid
3) It acts as a Lewis acid
4) All of the above

7. Which of the following statement is correct regarding inorganic Benzene?

1) Borazine is known as inorganic Benzene
2) Inorganic Benzene is more reactive than organic benzene as its bonds are polar
3) Both (1) & (2)
4) None of these

8. In structure of diborane, each Boron atom uses ____ hybridization for bonding.

1) sp^2

2) sp^3
3) sp^3d
4) sp^3d^2

9. Type here The exact chemical formula of Borax according to structures is
1) $Na_2\left[B_4O_4(OH)_4\right].8H_2O$
2) $Na_2\left[B_4O_5(OH)_4\right].8H_2O$
3) $Na_2\left[B_4O_5(OH)_4\right].7H_2O$
4) $Na_2\left[B_4O_5(OH)_4\right].6H_2O$

10. There is a considerable increase in covalent radius from C to Si and then after from Si to Pb a small increase in radius is observed. The correct reason for this is:
1) Effective nuclear charge increases down the group
2) Due to the presence of completely filled d and f orbitals in heavier member
3) Both 1 & 2
4) None of these

11. The correct order for electronegativity of group 14 element is
1) C > Si > Ge > Sn > Pb
2) C> Ge > Sn > Si > Pb
3) C> Si > Sn > Pb > Ge
4) C > Si = Ge = Sn = Pb

12. With respect to group 14 elements, which of the following is correct?
1) Group 14 contain metals, metalloid and non-metals
2) Melting and boiling point of group 14 element are much higher than those of corresponding group 13 elements
3) All member of group 14 are solids
4) All of these

13. Given reaction is

$Sn + 2H_2O \xrightarrow{\Delta} X + 2Y$ X and Y are
1) SnO, H_2
2) Sn_2O, H_2
3) SnO_2, H_2
4) Sn_2O_2, H_2

14. PbI_4 does not exist, the correct reason for this is
1) PbI_4 have very high lattice enthalpy
2) PbI_4 has high hydration enthalpy
3) Pb-I bond does not release enough energy to excite electron
4) Both 1 & 2

15. The crystalline form of carbon is
1) Diamond & Graphite
2) Fullerenes
3) Graphite & fullerenes

4) Both 1 and 2

16. Diamond is covalent in nature, yet it has high melting point why?

1) C-C bond length is very high
2) C-C bond is polar in nature
3) Diamond has a 3-D network involving strong C-C bonds
4) Both 1 &2

17. Which is thermos dynamically most stable allotrope of carbon?

1) Diamond
2) Graphite
3) Fullerenes
4) None of these

18. Fullerenes are pure form of carbon, the reason for this is

1) They have smooth structures without having dangling bonds.
2) They have rough structure with dangling bonds
3) They have slippery structure with dangling bonds
4) Both 2 & 3

19. Which of the following is true regarding Buckminster fullerene?

1) It contains 20 six-membered rings & 12 five-membered rings
2) A 6-membered ring is fused with 6 or 5 number rings
3) A 5-membered ring is fused only with 6 membered rings
4) All of these

20. Which among the following is crystalline form of silica?

1) Quartz
2) Cristobalite
3) Tridymite
4) All of these

21. Given reaction are

$SiO_2 + 2NaOH \rightarrow X + H_2O$

$SiO_2 + 4HF \rightarrow Y + 2H_2O$

X & Y are

1) $NaSiO_2, SiF_4$
2) Na_2SiO_3, SiF_2
3) Na_2SiO_3, SiF_4
4) Na_2SiO_4, SiF_4

22. Nitrogen found in earth's crust as-

1) $NaNO_3$ – chile salt petre
2) KNO_3 – Indian salt petre
3) $LiNO_3$
4) Both (1) and (3)

23. Which of the following is correct order of ionization Enthalpy?

1) N > As > P > Sb > Bi
2) N > P > Sb > P > Bi

3) N> P > As > Sb > Bi
4) N > P > As ≃ Sb > Bi

24. Which of the following correct order of Electronegativity?
1) $N < P < As > Sb > Bi$
2) $N > P > As > Sb \simeq Bi$
3) $N > P > As > Sb > Bi$
4) None of the above

25. Which of the following is not correct statement about physical properties of Nitrogen family?
1) All members are poly atomic solid but nitrogen is diatomic gas.
2) Metallic character increases down the group.
3) Boiling point decreases from Top to Bottom.
4) Melting point first increases up-to Arsenic & then decreases up-to Bismuth

26. HNO_2 disproportionate in acidic medium to give:
1) $HNO_3, NO_2 \ \& \ H_2O$
2) $HNO_3, N_2O_5 \ \& \ H_2O$
3) $NO_2 \ \& \ N_2O_5$
4) $HNO_3, NO \ \& \ H_2O$

27. Which of the following molecule forms $P\pi - P\pi$ multiple bond?
1) $N \equiv N$
2) $R_3P = 0$
3) $R_3P = CH_2$
4) All of the these

28. The correct order of reducing nature of group is hydride is:
1) $NH_3 > PH_4 > AsH_3 > SbH_3 > Bi\,H_3$
2) $NH_3 > PH_3 > AsH_3 < SbH_3 < BiH_3$
3) $NH_3 < BiH_3 > SbH_3 < AsH_3 < PH_3$
4) $BiH_3 > SbH_3 < AsH_3 < PH_3 < NH_3$

29. What are two Impurities are formed during production of N_2 by treating an aqueous solution of NH_4Cl and $NaNO_2$
1) NO_2 and HNO_3
2) NO and HNO_3
3) NO and HNO_2
4) N_2O and HNO_3

30. What is X, Y and Z in the given reaction respectively

$$NH_2CONH_2 + 2H_2O \rightarrow (NH_4)_2CO_3 \rightleftharpoons X + Y + Z$$

1) N_2, H_2O, CO_2
2) NH_3, H_2O, CO_2
3) N_2, H_2, CO_2
4) NH_3, H_2, CO_2

Answer Key

1	2	**16**	3
2	4	**17**	2
3	2	**18**	1
4	1	**19**	4
5	3	**20**	4
6	4	**21**	3
7	3	**22**	4
8	2	**23**	3
9	2	**24**	2
10	2	**25**	3
11	4	**26**	4
12	4	**27**	1
13	3	**28**	2
14	3	**29**	2
15	4	**30**	2